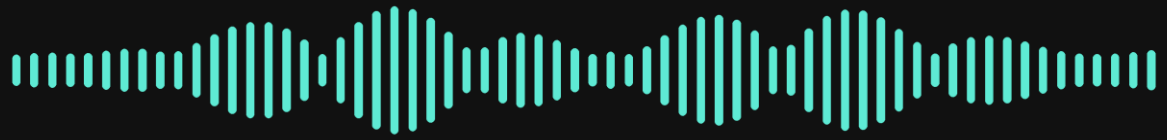
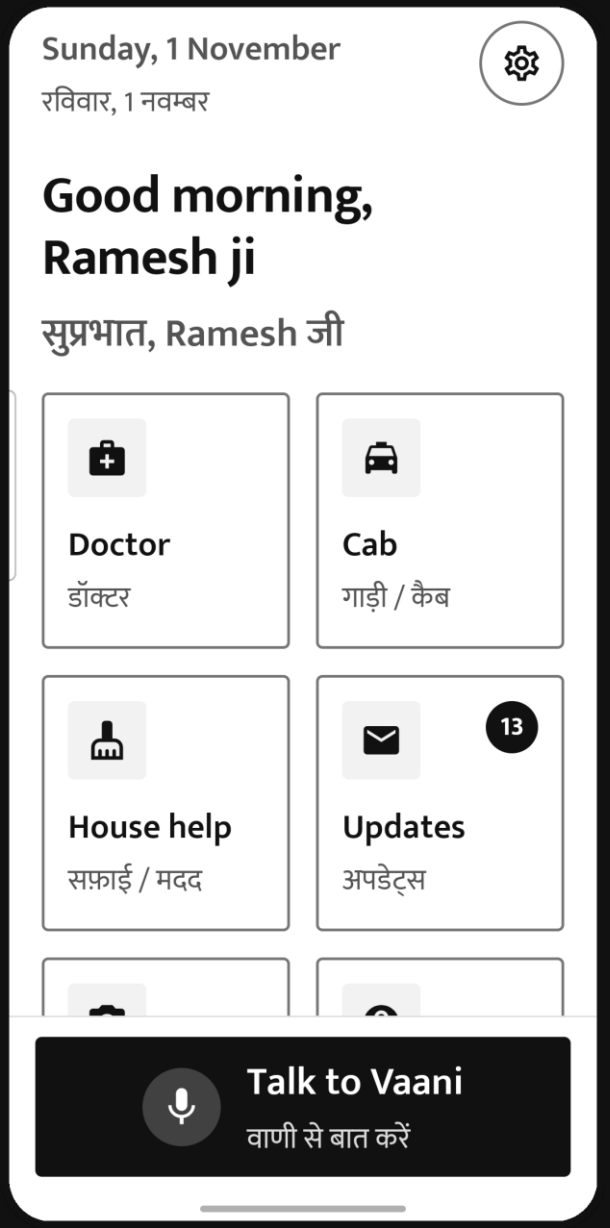
Vaani

वाणी

A voice companion that uses Android for people who find phones hard

Speak in Hindi, English or Hinglish. Nothing irreversible happens until you say “haan”.





THE PROBLEM

Phones now run daily life, and many people are left out

Cabs, doctors, bills and bank OTPs all arrive through apps built for young, literate, sighted users

149 million

Indians aged 60+ in 2022 (10.5%). Projected to be 20.8% of India, 347 million people, by 2050.

42%

of people aged 70+ have smartphone access. Only about two in five people over 60 can use the internet.

₹1,935 cr

lost to “digital arrest” scams in 2024, across 1,23,672 cases, nearly triple the cases of 2023.

Today they hand the phone, and the OTP, to whoever is nearby, go without, or get scammed. Voice assistants can’t finish a task in another app; screen readers can’t complete a booking.

Sources: India Ageing Report 2023, UNFPA & IIPS (via Down To Earth); NSS CMS:T 2025 analysed by LEAD at Krea University (ThePrint); MHA written reply, Rajya Sabha, 12 Mar 2025 (PTI)

WHO IT IS FOR

Built around three very different people

Same app, three different assistants: language, memory, appointments and skills all change with the person

R

Ramesh Sharma, 74

Hindi only · type 2 diabetes, knee arthritis

Reports his sugar every morning before chai. Son Amit lives in Pune.

S

Sunita Verma, 67

Hinglish · high blood pressure

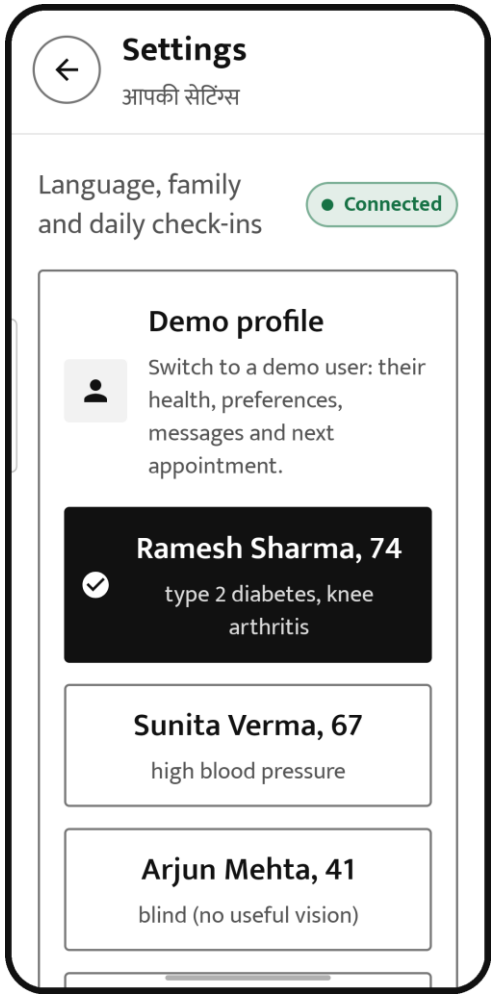
Checks BP after her evening walk. Loves achaar, asked Vaani to watch her salt.

A

Arjun Mehta, 41

English · totally blind, uses TalkBack

Every number must be spoken. Pays by UPI, splits the bill after Friday dinners.



Real app: Settings → Demo profile

HOW IT WORKS

The AI suggests. The phone’s code decides. You say “haan”

The rules live in the phone’s code, so they hold even if the AI is fooled



You speak
Hindi, English or Hinglish

Vaani understands
What you want, and where

AI picks one step
Tap, type or scroll

Code checks it
Rules the AI can’t override

Asks “haan?”
Before anything final

Acts in the app
Then looks again

A yes before anything final

Book, pay, send and add to cart wait for your yes.

Stops at secret screens

PIN, OTP, password and login screens stop it.

A short list of apps

Never banking, UPI, WhatsApp or email.

Only links it built

The AI can’t open a link it made up.

Hard limits

25 steps at most; “Stop” works at any moment.

Screen text is data

Orders hidden in a message are ignored.

USING APPS FOR YOU

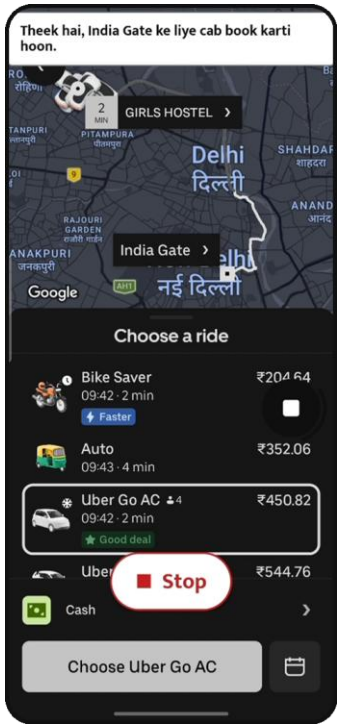
A real run: “Yahan se India Gate tak cab book karein”

Captured from our run on a phone with the real Uber app



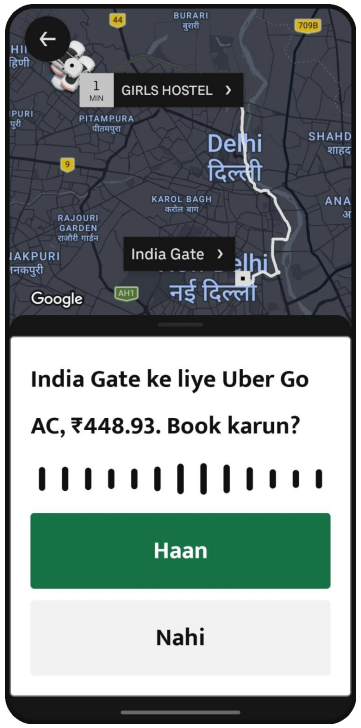
1 It listens

Vaani repeats the request back in the user’s own words.



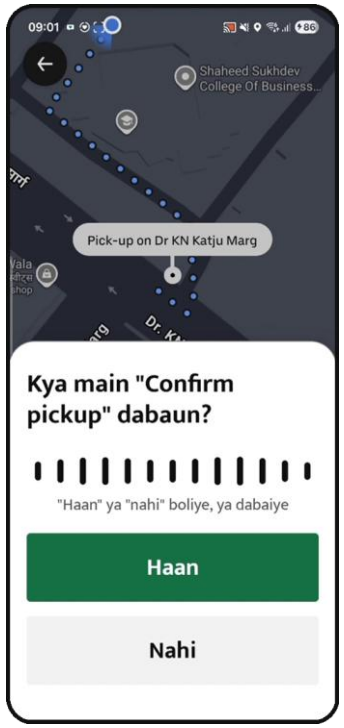
2 It picks a car

“Cab” means a car, so it chooses Uber Go AC, not the cheaper bike.



3 It asks first

Ride and fare read aloud. Only “haan” or a tap books it.



4 Again at pickup

Confirming the pickup spot is also a yes/no question.

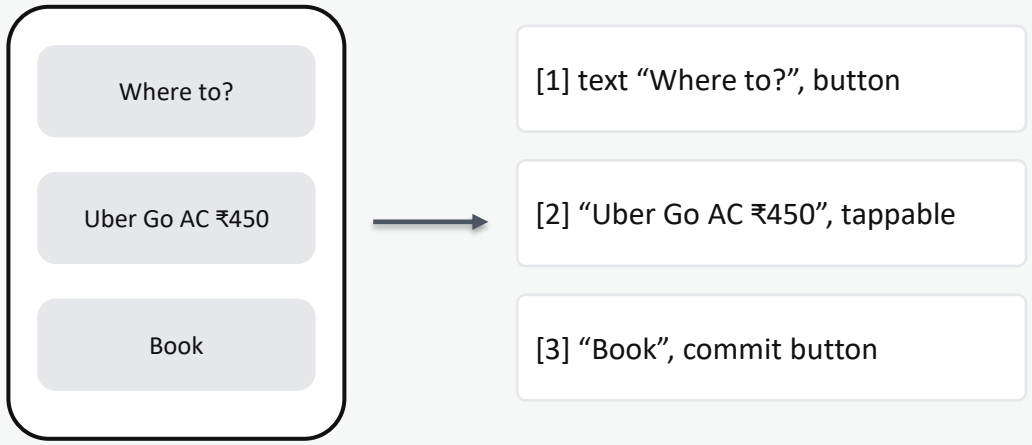
USING APPS FOR YOU

Two kinds of apps, one set of rules

Vaani uses the same screen a person sees, read through Android's accessibility service

Uber: the screen describes itself

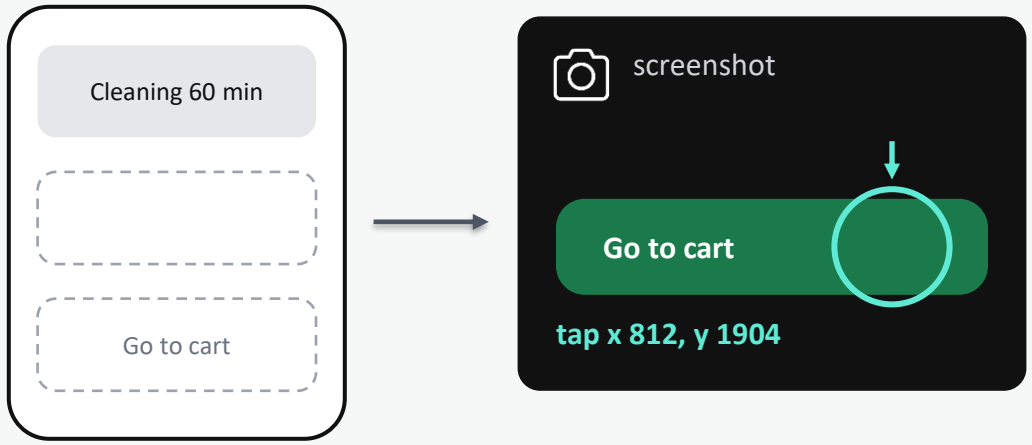
Vaani reads every button as text



Uber. Vaani's own code builds the deep link with the destination set; the AI never invents links. Taps go to named elements, so a routine can later replay them.

Pronto: parts of the screen are hidden

Vaani adds a screenshot and taps by position

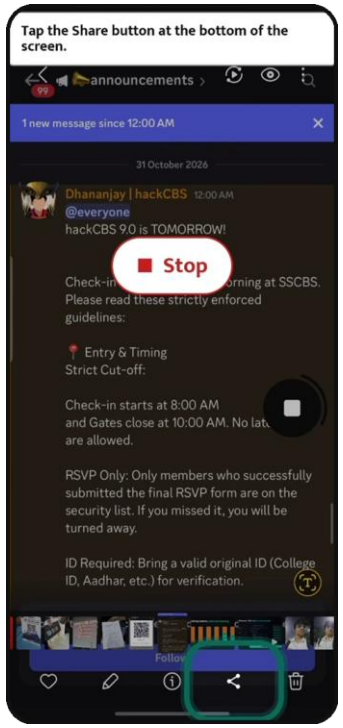


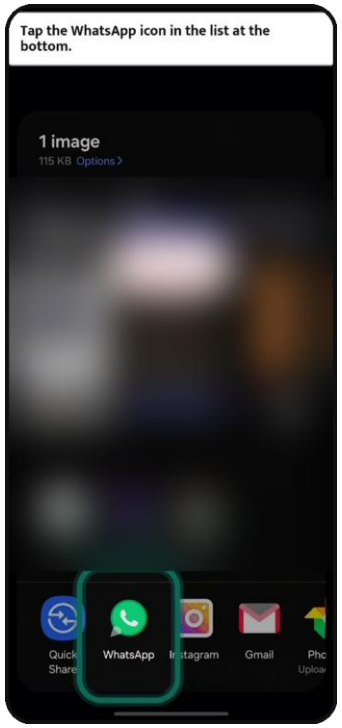
Pronto. "Added" and the "Go to cart" bar never reach accessibility, so the AI reads the picture. Code still needs a screenshot before any position tap, and Vaani stops at the cart.

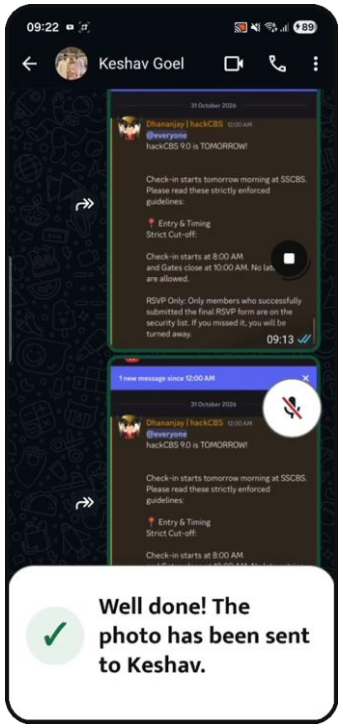
USING APPS FOR YOU

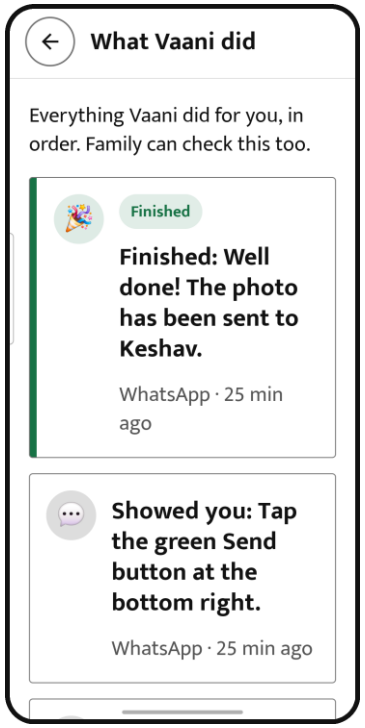
“Show me how”: Vaani points, you tap

A real run: “I want to send this photo to Keshav.” Vaani never taps in this mode









1. **Rings the button**
   “Tap the Share button at the bottom of the screen.”
2. **One step at a time**
   Waits for the user’s tap, then rings the next one.
3. **Praises the finish**
   “Well done! The photo has been sent to Keshav.”
4. **Family can check**
   Every step is in “What Vaani did”, in plain words.

HEALTH

Doctor: is this an emergency? Checked on every sentence

A rule-based safety net runs before any AI, so a slow model never delays help

You describe it
In your own words and language

Safety net, no AI
16 red flags in 3 languages; understands “paseena nahi hai”

Emergency

Siren and a 15-second countdown; “I’m OK” cancels

Ambulance request with priority, life-support level, GPS and a paramedic handover

Live map with ETA. Never dials 108 by itself

Not an emergency

An AI nurse asks at most 3 questions, red flags first

It must answer YES, NO or UNCERTAIN; uncertain is treated as an emergency

Then: find the right hospital



Real app: the doctor screen listens first

India’s 108 has no public API, so dispatch goes to our own realistic ambulance API stand-in, built so a real provider can be plugged in.

HEALTH

Picking a hospital, then checking with a real person

Facilities, urgency, fee limit, insurance and hospitals to avoid all come from the conversation



Find

Real places from Google, widening 8 → 15 → 25 km, with real driving times.

Filter

Drop no phone, closed now, the wrong kind, and hospitals you said no to.

Score

Can they treat this need, how near, how well reviewed; urgency changes the weights.

Call, and book while they hold

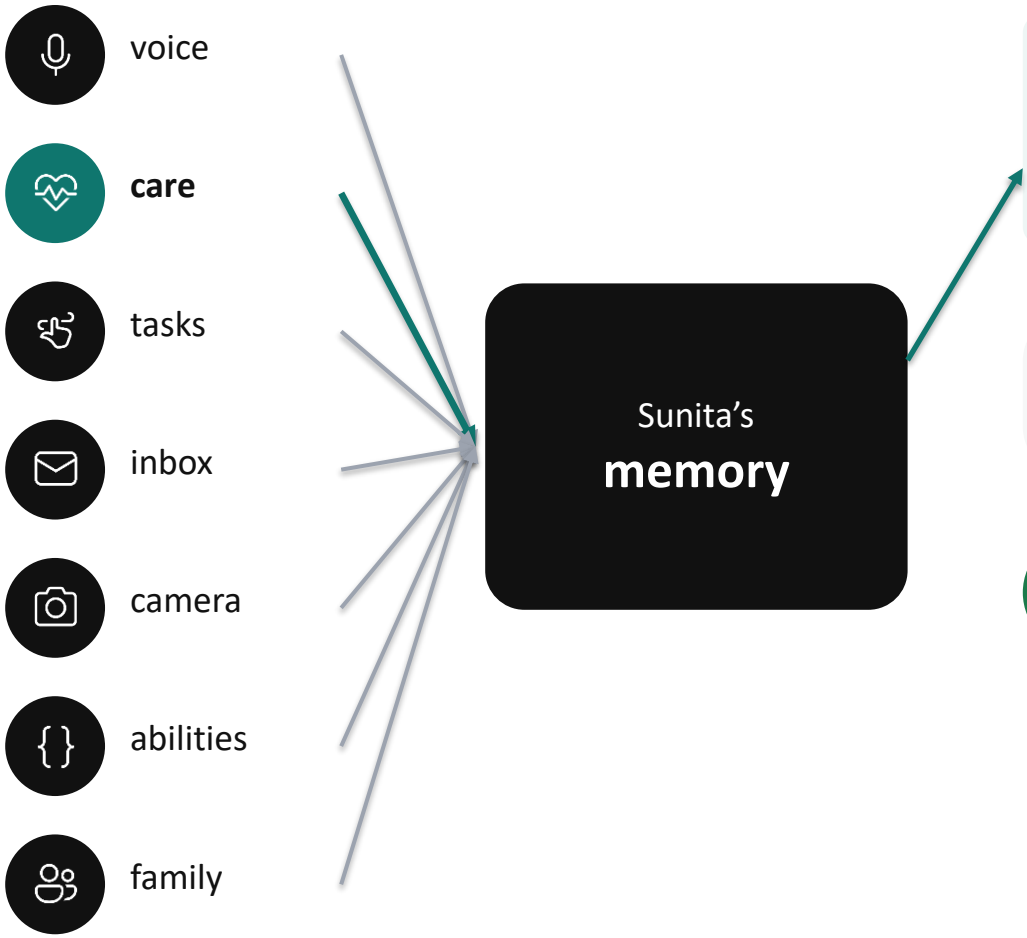
The top 3 are phoned at once to confirm the specialist, facilities, fee and slots. On your “haan”, it books on that same call.

Demo safety: the hospitals shown are real, but every number is swapped for our own test phone. Only allow-listed numbers can ever be dialled.

PERSONAL

One memory per person, written by seven agents

Each fact records which part of Vaani learnt it, and where, in that person's own Meko datapack



Learnt by the care agent *"while the hospital was on the line"*

Appointment with Dr. Rakesh Gupta at Fortis Memorial Gurugram, Sector 44, Tue 3 Nov, 11:00 · address and map location attached

Days later: "Mere doctor appointment ke liye cab book kar do"

The cab goes to Fortis, Sector 44.

She never said where. Vaani says the destination before booking, and it still works after a reinstall or on a new phone.

PERSONAL

Each person grows different skills

The code keeps the numbers right; the AI words the reply in that person's own style

Ramesh

Sugar diary checked against Dr. Kapoor's targets

Metformin countdown: tells Amit when to reorder

Shows a mango drink to the camera



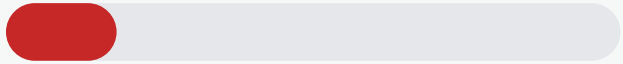
≈ 7 teaspoons of sugar in this pack

Sunita

BP diary with a 7-day average; 180/120 is always urgent

Salt budget against WHO's 5 g a day

Shows a packet of chips



0.9 g salt: 18% of her day

Arjun

Spoken expense log by day, week or category

Bill splitter: ₹2,350, 4 people, 10% tip → ₹647 each

Shows either pack



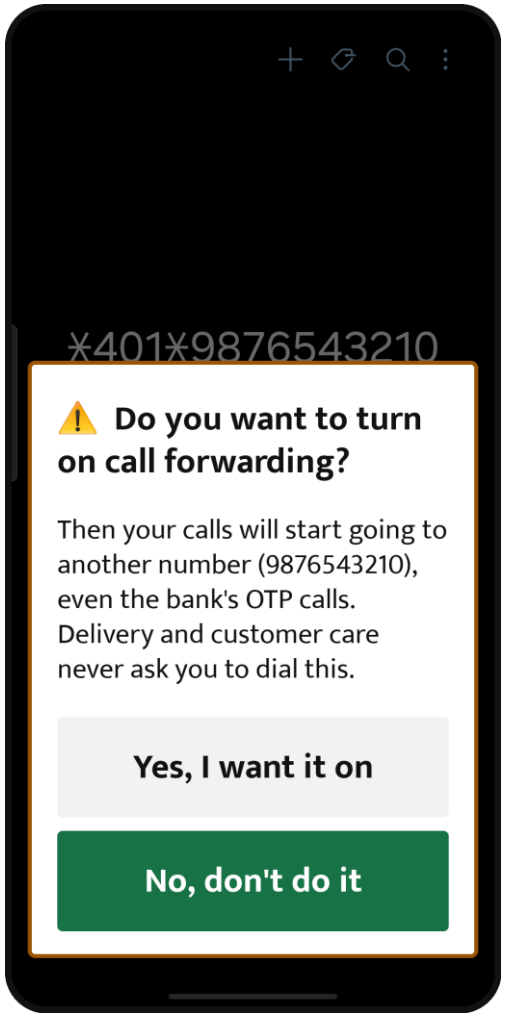
Just the label, read aloud

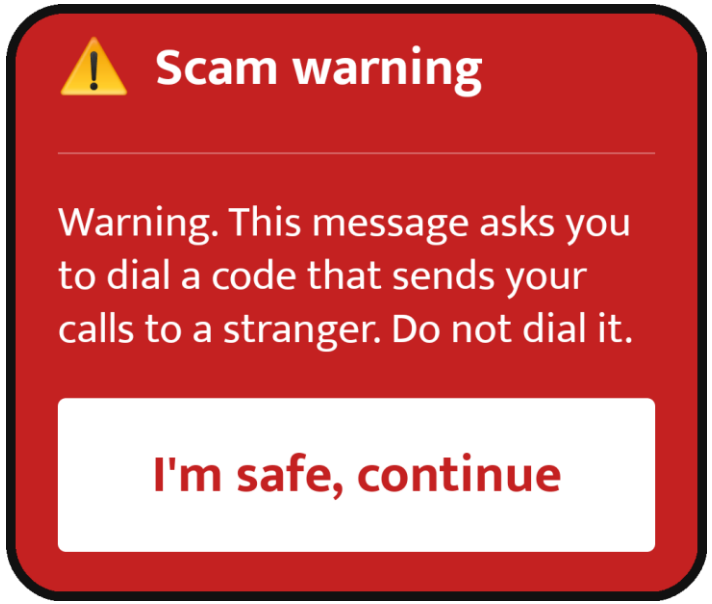
Urgent results (sugar under 70, BP 180/120) are spoken exactly as the code wrote them, without waiting on a model.

TRUST

Scam Shield warns before money moves

Real screens from the app





Checked on this phone

Scam Shield runs on the device. OTPs never leave it, even when a warning is raised.

Call-forwarding codes

A message asking you to dial *401* gets a red warning. Typing such a code in the dialler asks first.

During a call with a stranger

If an OTP arrives, or a payment or screen-sharing app opens, a loud warning plays in the earpiece.

Links and fake banks

APK links and lookalike bank sites are flagged; links open in our sandbox browser, never on the phone.

Never waits on the internet

Safety voice clips are bundled in the app.

TRUST

Privacy, measured on our own test set

An audit script runs these checks, and the same checks run as automated tests

Nothing private left the phone

We passed 12 realistic SMS, email and WhatsApp messages holding 22 private values (OTPs, account, card, Aadhaar and PAN numbers, UPI IDs, amounts) through the real notification path. None appeared in what would be sent.

Nothing personal in server logs

We sent 32 requests packed with names, phone numbers and the same secrets through every server route and captured all logs at debug level. None of those values appeared.

Only the owner’s phone can read

Each install holds its own secret; the server keeps only its hash. Every forged or cross-device request we tried was rejected.

Links checked without personal parts

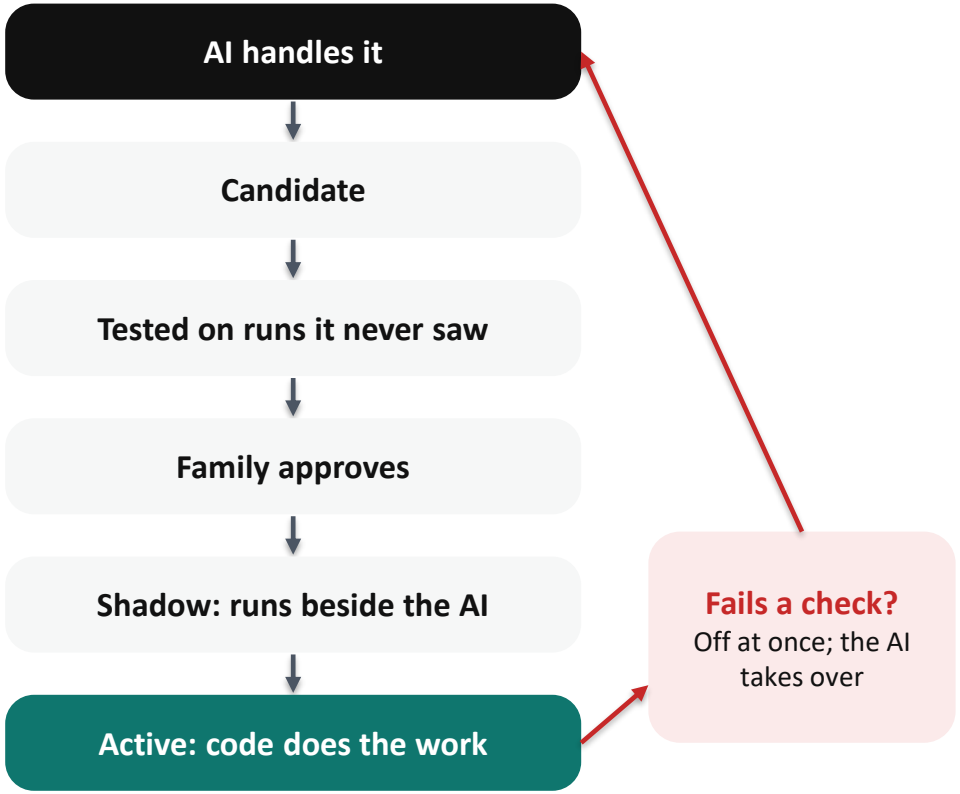
Query strings, tokens and IDs are stripped before a link is checked. All 10 of our test links came out exactly as expected.

How: OTPs and messages from saved contacts never leave the phone. Other private values become placeholders like [P12], and the phone reads the real value aloud itself. The audit also found two gaps; both are fixed.

UNDER THE HOOD

Repeated work becomes tested, sealed code

Routines, recognisers and personal abilities all follow this loop; the AI still writes every reply



Our DigitalOcean server: every layer, every run

Static check: safe imports only; no files, network, eval or exec

Fingerprint: SHA-256 checked before every run

Docker container per job: no network, read-only, 128 MB

Limits: 10 s per case, checked output

`run(job)`

~0.55 s
per sandboxed run

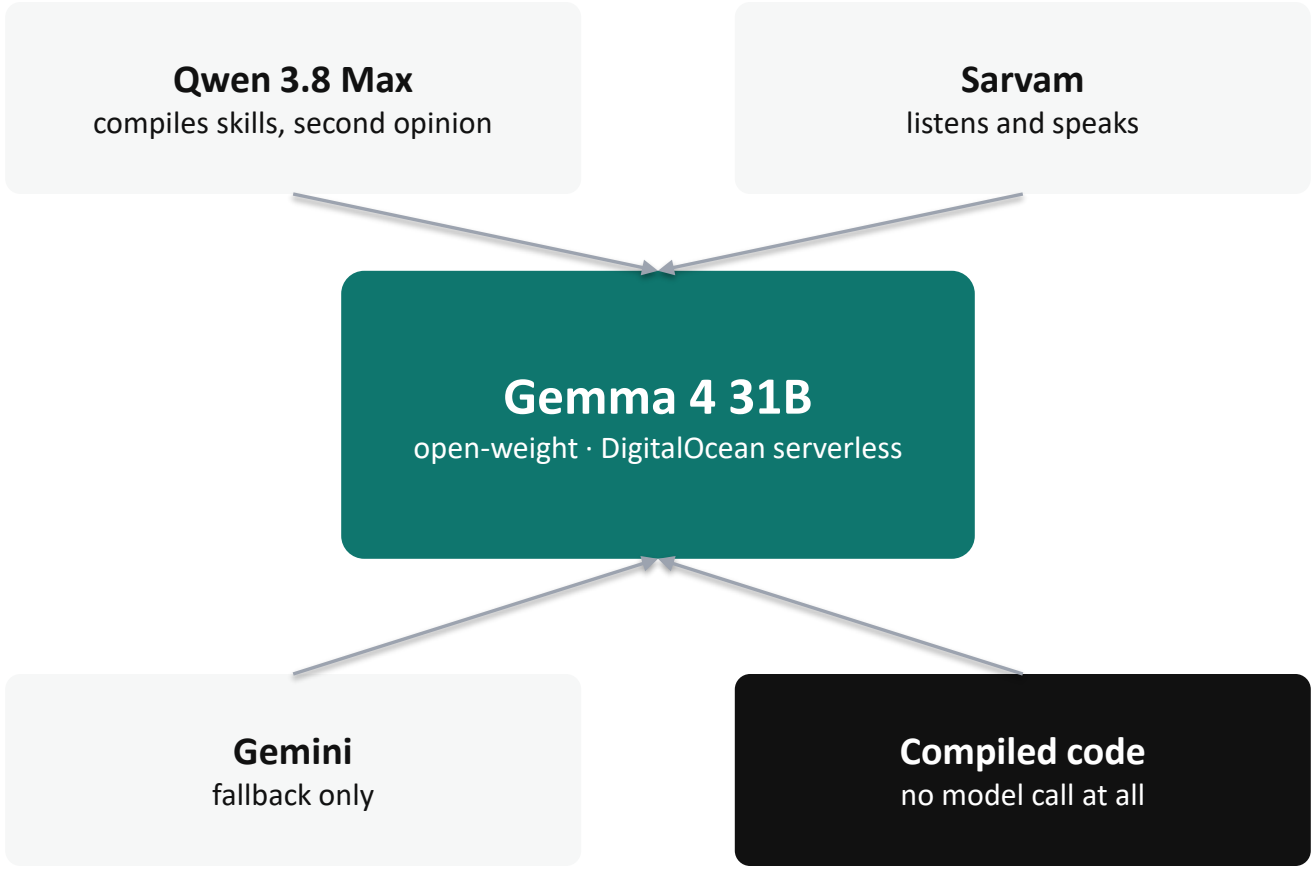
0.7 s
to kill a test memory bomb

Any error
and the AI answers instead

UNDER THE HOOD

Open models, and less AI over time

Every model choice lives in one server file, so providers change without an app update



What keeps costs down

Repeated work becomes code, with no model call

Simple cab requests are understood on the phone

Screenshots only when needed, and compressed

An on-phone classifier when the network is down

Every feature runs without API keys in mock mode

~55,000

model tokens per person per month (simulated estimate)

38%

fewer AI steps once the cab routine is learnt

Your codes never leave your phone. Nothing happens until you say “haan”.



github.com/Aryannnthakurrr/Vaani

Aryan Thakur · Keshav Goel · Anish Manchanda

